# DunesForge - wind, sand, snow and water that behave

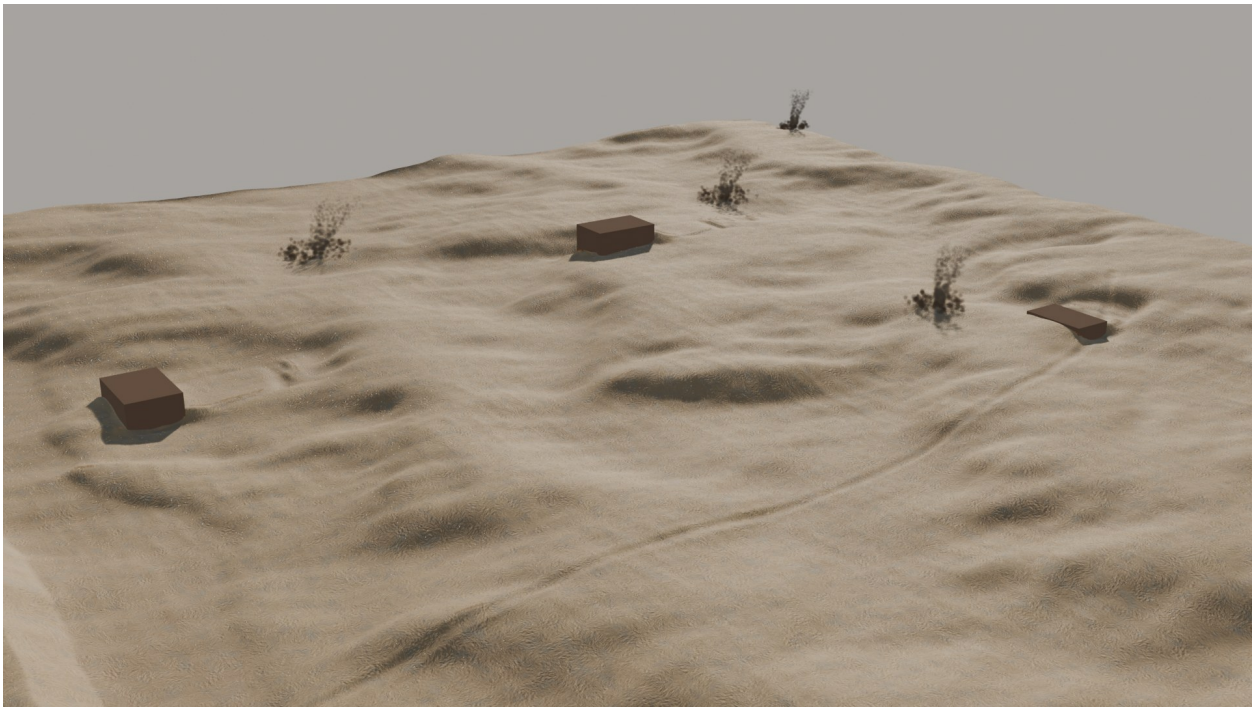
**A physically based dune, snow, water and erosion simulator that runs inside Blender, on your GPU, in the viewport.**

## Start here

| I want to... | Go to |
|---|---|
| **Install it and see something move in 5 minutes** | **-> QUICKSTART.md** |
| **Install it, step by step, with troubleshooting** | -> INSTALL.md |
| **Look up what one specific button does** | -> REFERENCE.md |
| **Follow a guided tutorial** | -> QUICKSTART.md § Tutorials |
| **Just open a finished scene and take it apart** | -> QUICKSTART.md § The 66 demo scenes |

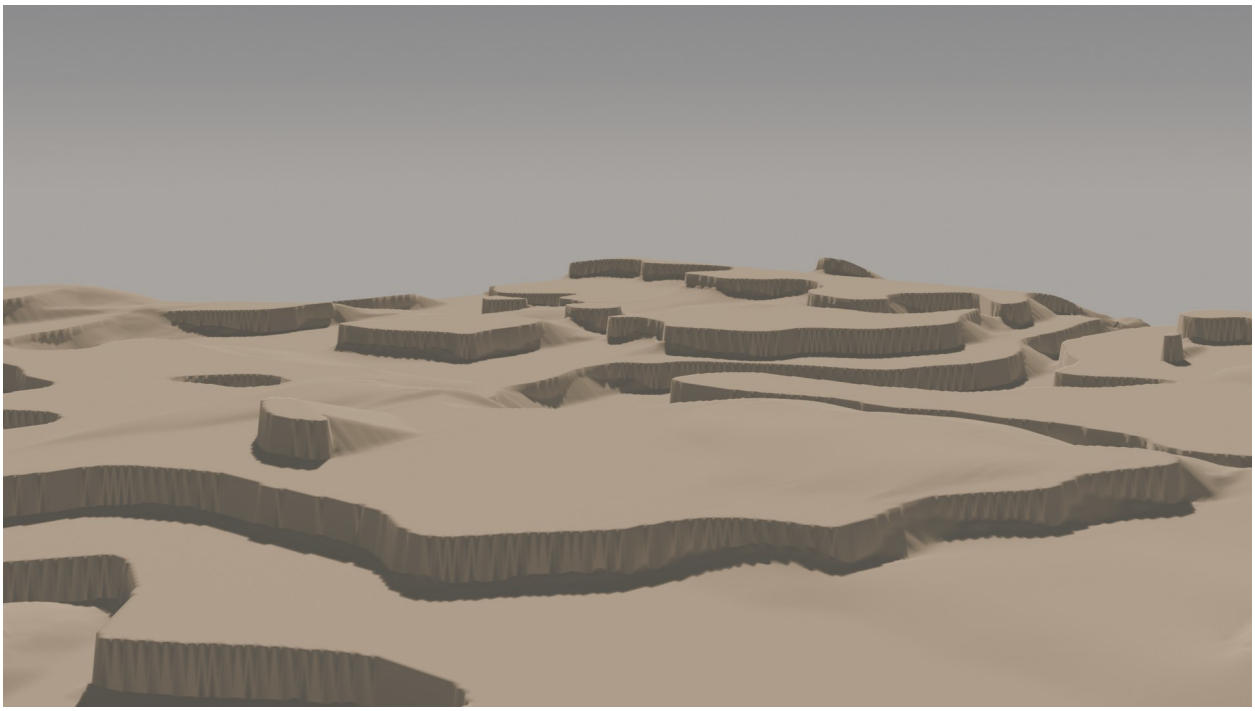
**A note on the name.** The product is **DunesForge**. Inside Blender the add-on's panel is still labelled **TyDunes** - that is the tab to look for under **[N]**, and the file you install is `tydunes_blender.zip`. The interface will catch up with the name; nothing about how it works changes.

## What it looks like



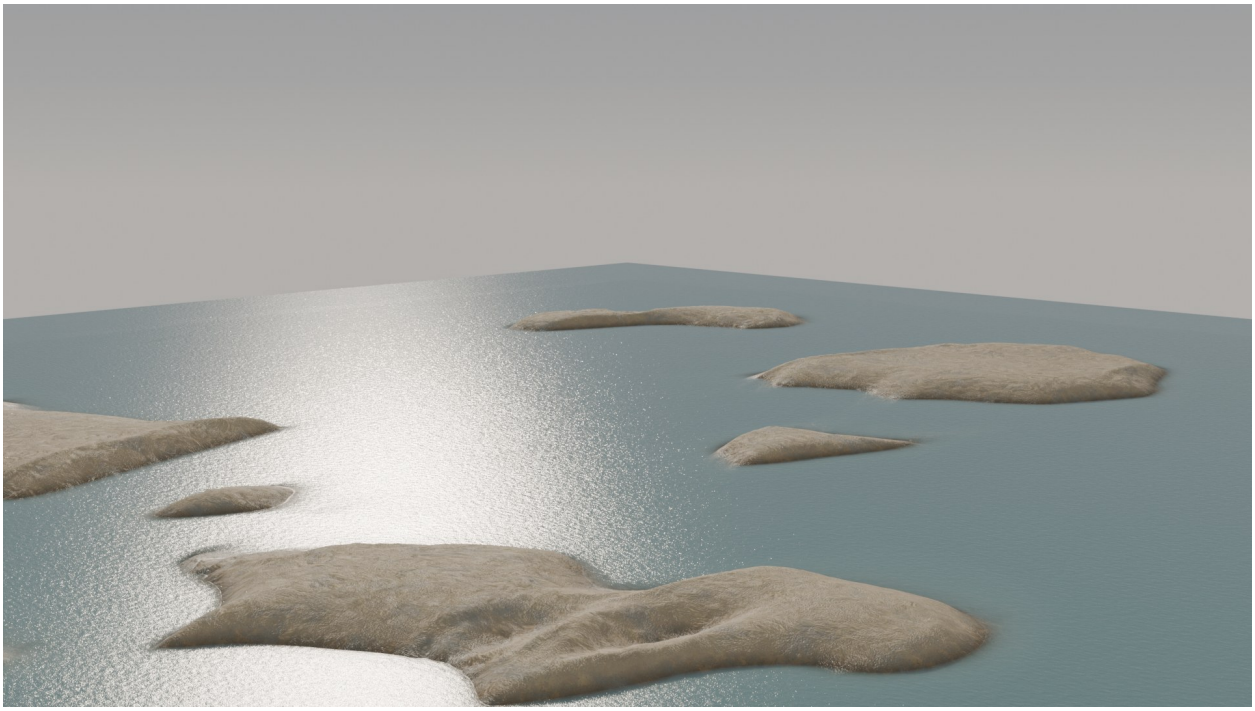
*A 400 m desert: dunes, shelters, vegetation and vehicle tracks, all simulated*

Grand Scene: Desert Expedition - one of 66 demo scenes that ship with the add-on. Everything you see here is simulated, not sculpted.



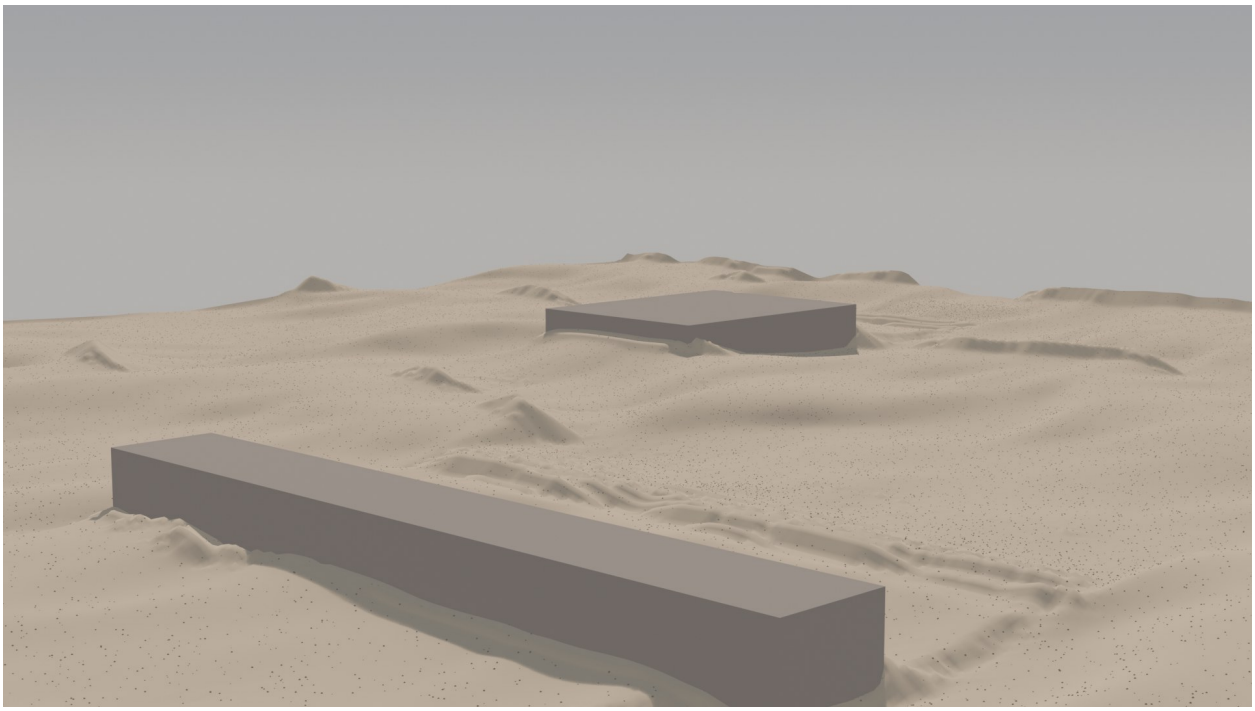
*Sand over a layered mesa: it settles on the shelves and in the hollows, and leaves the steep walls bare*

Terrain: Mesa - sand collects where it physically can. The walls stay bare because they are steep AND high at once.



*Ice floes on open water with a sun glint*

Ice: Pack, Ridges and Leads - the same solver, a different material model.



*Dunes forming around two rock blocks, with wind shadows and lee tails*

Desert: Dunes + Obstacles - move a rock while it plays and the wind shadow follows it.

## What this is, in one paragraph

Point DunesForge at a terrain, give it a wind direction, and press Play. Sand starts moving the way sand actually moves: it climbs the windward slope, spills over the crest, and slides down the lee face when the pile gets too steep. Do it long enough and you get real dunes - barchans with horns, transverse ridges, star dunes - in the shapes the wind and the sand supply dictate, not shapes you sculpted. The same solver drives snow drifts, slab avalanches, flowing water, rain, frost, vegetation and wind erosion.

It is a Blender add-on. Nothing is exported, nothing round-trips to another application, and the result is an ordinary Blender mesh you can render, shade, cache and hand to anyone.

## The problem it solves

**Sculpting a desert by hand is guesswork, and it shows.** Dunes have a signature: a long gentle windward slope, a sharp crest, and a lee face sitting at the angle of repose - about 34 deg for dry sand, and the same on every dune in the shot because it is set by physics, not by taste. Audiences do not know the number, but they recognise when it is wrong. Getting it right by hand across a whole landscape is days of work, and any change to the wind direction means doing it again.

**Scattering assets does not fix it either.** A library of dune meshes gives you dunes that do not interact - they do not shelter each other, do not bury the rock you placed between them, and do not respond when you move anything.

**DunesForge gives you the process instead of the product.** You set the conditions; the shapes follow from them. Change the wind by 30 deg and the whole field reorganises. Drop a boulder in and sand piles up in front of it and tails off behind it. Add plants and they hold the sand down until they get buried.

### *Things that are genuinely hard to fake, and come free here*

responds to it - drifts against walls, tails behind rocks, scour pits at corners.

barchans that march; a deep bed makes connected transverse ridges. Same solver, different supply.

layers, and releases as slabs that scour their own path.

resists erosion, and the channel undercuts its own banks.

- **Sheltering and interaction.** Every obstacle casts a wind shadow, and sand
- **Sand-starved versus sand-rich terrain.** A thin veneer over bedrock makes
- **Snow that is not just white sand.** Snow sinters, hardens, forms weak
- **Water that talks back to the ground.** Flowing water wets the sand, wet sand
- **Time.** Every one of these is an animation, not a still. Your dunes migrate.

## What we are good at

**The physics is the research model, not a look-alike.** The wind field uses Jackson-Hunt shear perturbation with lee-side flow separation, and the sand relaxes by avalanching at the repose angle - the Kroy / Sauermann / Durán lineage that the scientific literature on dune modelling is built on. Where a parameter has a meaning in that literature, the control carries that meaning.

**Every physical feature is off by default and identity at its default value.** Turning a feature on is a decision you make; leaving it alone means the solver runs exactly the arithmetic it ran before that feature existed. Scenes do not change under you when the add-on is updated.

**Claims in the interface carry measurements.** Where a tooltip says a feature does something, that came from a measured A/B run against a control, not from an intention. Where a feature turned out not to earn its place in a scene, it was removed rather than shipped with a promise attached.

**It runs on the GPU and degrades gracefully.** CuPy on an NVIDIA CUDA card when you have one, and a fully vectorised NumPy CPU path when you do not - same results, different speed.

**66 finished demo scenes ship with it.** Not screenshots: actual scenes, built by an operator, that you can open, run, take apart and steal settings from. Every major feature has at least one scene that exists to demonstrate it.

## Who it is for

eroded badland that holds up in a wide shot.

a creature, a building, a wind change.

running a DEM solver.

importable and exportable, and the behaviour documented with numbers.

- **Environment artists** who need a desert, a snowfield, a frozen coast or an
- **VFX and previs** work that needs sand or snow to *respond* - to a vehicle,
- **Motion designers** who want the look of real granular physics without
- **Technical artists and TDs** who want the parameters exposed, the maps

You do **not** need to understand aeolian physics to use it. The presets and demo scenes are there precisely so you can start from something that already works.

## What you need

| | |
|---|---|
| **Blender** | 4.2 or newer (tested through 5.3) |

| | |
|---|---|
| **OS** | Windows (primary), Linux |
| **GPU (recommended)** | NVIDIA with CUDA, via CuPy - this is what makes it interactive |
| **GPU (optional)** | No CUDA card? The NumPy CPU path runs the same simulation, slower |
| **RAM** | 8 GB is workable; 16 GB is comfortable at higher grid resolutions |

> **On speed:** simulation cost scales with grid resolution, and the resolution is yours to pick. The **Performance** panel exposes the trade-off directly, and every scene ships at a resolution that runs interactively on a mid-range CUDA card. See REFERENCE.md § Performance.

---

## How the documentation is organised

**QUICKSTART.md** - read this first. It is the handbook: your first dune in five minutes, how the simulator actually works, a tour of every panel, a working method from idea to render, **sixteen step-by-step tutorials**, all 66 demo scenes grouped by what they show, a glossary, a troubleshooting section and checklists.

**REFERENCE.md** - every control in every panel, what it does, what it defaults to, and when you would reach for it. Organised the way the interface is organised, so you can find a control by where you saw it.

**INSTALL.md** - installation in detail, including the GPU setup, what to do when CuPy is missing, and the common failure messages.

---

*DunesForge is a faithful rewrite of the 3ds Max / tyFlow DunesForge solver, rebuilt as a native Blender add-on.*